Algorithm 7: Greedy-stochastic rollout probe

Input:

- L_{start}. The starting label $(v, t, C, \text{pred}, \text{start\_time})$
- E_s. Set of scheduled edges $(v, u, \text{dep}, \text{arr})$
- E_t. Set of transfer edges (v, w, Δ)
- country(v). Country associated with node v

Output:

- L_{final}. The final label reached when the 24-hour clock expires

```
L ← L_start
// Repeat until no more valid moves within 24 hours
repeat
    ValidSteps ← ∅
    // Collect valid scheduled edges
    for all (L.node, u, dep, arr) ∈ E_s do
        if dep ≥ L.t & arr − L.start_time ≤ 1440 then
            ValidSteps ← ValidSteps ∪ {(u, arr, country(u))}
    // Collect valid transfer edges
    for all (L.node, w, Δ) ∈ E_t do
        t_new ← L.t + Δ
        if t_new − L.start_time ≤ 1440 then
            ValidSteps ← ValidSteps ∪ {(w, t_new, country(w))}
    if ValidSteps = ∅ then
        break // No more moves possible within 24 hours
    // Assign greedy-stochastic weights to each valid step
    for all step s = (u, t_new, c_new) ∈ ValidSteps do
        Δt ← max(1, t_new − L.t) // Total wait + travel time
        if c_new ∉ L.C then
            s.weight ← 10000/Δt // Huge bias for unvisited countries
        else
            s.weight ← 10/Δt // Low bias for transit/re-entry
    // Pick one step randomly with probability proportional to its weight
    s_chosen ← WeightedRandomChoice(ValidSteps)
    C′ ← L.C ∪ {s_chosen.c_new}
    L ← (s_chosen.u, s_chosen.t_new, C′, pred = L, start_time = L.start_time)
until false
return L
```